Documentation - Power Electronics Onboarding

Activity 1 - Conceptual

- Ohm's Law: V=IR
- I (Current) = dQ/dt
- Series (addition) and parallel resistor laws($\frac{1}{\frac{1}{R1+R2}}$)
- P (power) = V*I = I^2*R
- Kirchoff Loop Rules -> sum of the voltage drops is 0
- Linear loss: (Vin - Vout) * I(current) -> When using a linear system to reduce from the high voltage of the bus, it burns off the linear power loss as heat, which is why we switch to buck converters, which allow us to reduce the voltage from ~13.5 V to 5V by flipping switches thousands of times. Switches are on 37% of the time (Duty cycle, Vout = D*Vin => 5/13.5 = 0.37), and the average of the voltage coming out after going through the pulses should be about 5V. The buck chip will adjust the duty cycle accordingly based on input voltage to hold 5V as much as possible.
- To smooth out the current, an inductor is introduced so that the current cannot fluctuate rapidly (only gradually decrease / increase), and a capacitor is also introduced to smooth out the voltage in the same way.
- To find the rating of the inductor, need to calculate ripple current and solve for the L in the equation:

$$L = \frac{(Vin - Vout) * D}{Inductor\ ripple\ current * Frequency\ of\ buck\ converter\ switching}$$

- Capacitor is sized using the following equation:

$$C = \frac{Inductor\ ripple\ current}{8 * Frequency\ of\ buck\ converter\ switching * Ripple\ Voltage}$$

Activity 2: Components Chosen / Justification

- **Buck IC:** Texas Instruments LMR51420XFDDCR; Datasheet: https://www.ti.com/lit/ds/symlink/lmr51420.pdf
    - Fits the Input Voltage requirements (4.5 - 36V range), Output Voltage (Adjustable, with a reference voltage of 0.6), Output Current >= 2A, Switching Frequency >= 500 kHz, and DigiKey distributed.
    - The cost is $1.58 per unit.
    - Another shortlisted alternative I had was the AP62300T Buck IC chip, and this is roughly $0.38 per unit. However, the maximum permissible voltage is much lower than the chip I chose, and does not leave a lot of range for any varying voltage supply. I figured as the bus mentioned is supposed to be for 12 V, but instead tends to give 13.5 V, spikes are likely possible and anything above 18 V (its max rating) would run the risk of frying this chip. Furthermore, the TI chip is rated for higher temperatures and can maintain a nominal frequency of 500 kHz comfortably even with light loads (the AP chip lowers its frequency as a power saving measure.)
    - Calculations behind choosing this chip:
        - **Minimum output power: P = Vout * Iout = (5 V) * (2 A) [**Required output voltage multiplied into the required output current, as found on the onboarding guide) => **10 W.**
        - **Required Vin max rating:** The maximum voltage rating should be greater than 1.5 times the maximum voltage noted in the onboarding guidelines (to meet the 50% margin noted on the Low Voltage standards document). **15 V * 1.5 = minimum 22.5V** I know that the Low Voltage 50% margin noted was for capacitors, but I figured that the same margin tolerances would affect the buck IC as well as it faces the same volatility from the bus, so I wanted to make sure the IC had the same margin range as the capacitors. Since the maximum voltage rating for the TI chip is 36 V, it is way higher than the 22.5V needed with the 50% margin factored in.
        - **Frequency:** This is the FPWM model. As stated before above, the datasheet guarantees that the chip will maintain its typical frequency even at low loads, and its typical frequency is 500 kHz.
        - **Feedback control:** The chip has two inbuilt systems, one to set the switches and one monitoring how much the output is, to adjust the duty cycle accordingly. The one monitoring the output will see the reference voltage (set to 0.6 volts) and compare that to a scaled down version of the output to see if they match. It will adjust the duty cycle accordingly for the switches in order to hold the output voltage steady at 5V. These were one of the major reasons I chose this chip, because these systems allow for faster reaction times and do not require extra external parts.

- **On-time/off-time**. Keeping in mind the maximum possible input voltage, the duty cycle becomes Vout / Vin = 5 / 15 = 0.33. The length of one switching cycle is 2 µs, which means that the minimum necessary time for the switch to be on is 0.33 * 2 = 0.66 µs. That also means that the shortest off-time necessary for the switch is: D = Vout / Vin = 5 / 10 (minimum range of input voltage), and therefore (1 - 0.5) * 2 µs = 1 µs. As the minimum on-time and minimum off-time is well below these ranges (70 and 150 ns each), this chip supports our shortest possible switch on-off pulse cycle. Also from the datasheet, I found that:
    - Minimum Vin without frequency foldback (reduced frequency) is

$$V_{IN_MIN} = \frac{V_{OUT}}{1 - f_{SW} \times T_{OFF_MIN}}$$

      and plugging in the numbers, we get around 5/(1-500000 * 150ns) = 5.4 V
    - Maximum Vin without frequency foldback is

$$V_{IN_MAX} = \frac{V_{OUT}}{f_{SW} \times T_{ON_MIN}}$$

      and plugging in the numbers, we get around 5/(500000 * 70ns) = 143 V.
    - The on-off times are denoted by T off min and T on min, and they show that within a range of 5.4 V to 143 V (although the chip is capped by a 36 V maximum input), we can hold the frequency at 500 kHz, which makes this chip ideal for a crucial step-down voltage component.
- **Inductor:** Eaton DR1040-100-R (10 µH, shielded version).
    - Cost: $0.78/unit.
    - **Calculations justifying inductance value:** Taking the highest possible input value, as the ripple current percentage should be the highest there. Duty cycle at this point would be 5V / 15V = 0.33. The output current requirement is 2A (from the onboarding document). Therefore, the inductance value follows the equation of L = (Vin - Vout)*D/(Change in inductor current * Frequency of switching). When plugging in numbers, it becomes $\frac{(15-5)*0.33}{(0.3*2)*500000} = 11.1\mu H$, which I

rounded down to around 10 µH as the buck IC datasheet also recommends it for an output of 5V at a frequency of 500 kHz. The change in inductor current (30% of 2A), specifically the percentage of 2A, was also an optimal design choice in the buck IC datasheet:

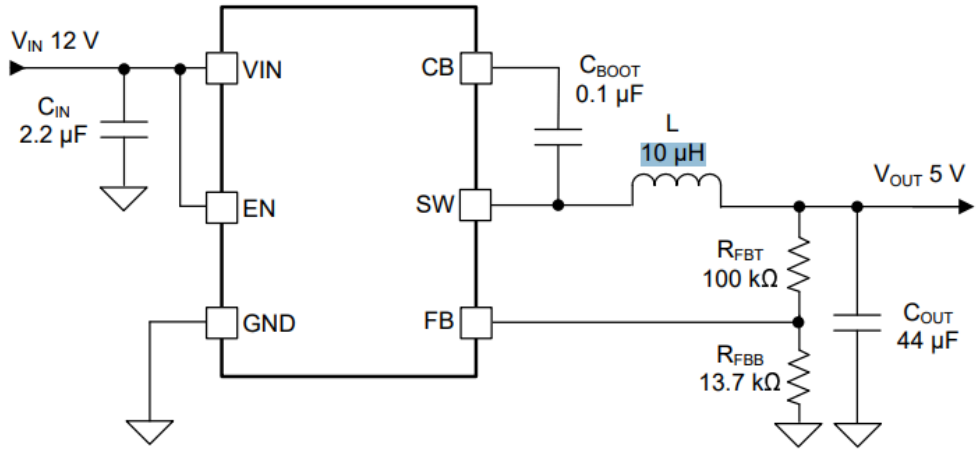
Equation 11 to calculate the minimum value of the output inductor. K_{IND} is a coefficient that represents the amount of inductor ripple current relative to the maximum output current of the device. A reasonable value of K_{IND} must be 20% to 60% of maximum I_{OUT} supported by the converter. During an instantaneous overcurrent operation event, the RMS and peak inductor current can be high. The inductor saturation current must be higher than peak current limit level.

$$\Delta i_L = \frac{V_{OUT} \times \left(V_{IN_MAX} - V_{OUT} \right)}{V_{IN_MAX} \times L \times f_{SW}} \quad (10)$$

$$L_{MIN} = \frac{V_{IN_MAX} - V_{OUT}}{I_{OUT} \times K_{IND}} \times \frac{V_{OUT}}{V_{IN_MAX} \times f_{SW}} \quad (11)$$

In general, choose lower inductance in switching power supplies because it usually corresponds to faster transient response, smaller DCR, and reduced size for more compact designs. Too low of an inductance can generate too large of an inductor current ripple such that overcurrent protection at the full load can be falsely triggered and generates more inductor core loss because the current ripple is larger. Larger inductor current ripple also implies larger output voltage ripple with the same output capacitors. With peak current mode control, ensure there is an adequate amount of inductor ripple current. A larger inductor ripple current improves the comparator signal-to-noise ratio.

For this design example, choose K_{IND} = 0.3. The minimum inductor value is calculated to be 9.7 µH. Choose the nearest standard 10-µH ferrite inductor with a capability of 3-A RMS current and 4-A saturation current.



- A cheaper version of this that fit the buck IC's optimal specifications did exist (Bourns SRN6045TA), however this was not fully shielded and any residual magnetic fields could affect the operation of the other components due to noise.

- **Ripple Current Check:** The buck IC datasheet also specifies a 20% to 60% range for the ripple current ratio, and the inductor matches this specification. As its rating is 10 µH, the equation is:

$$\Delta i_L = \frac{V_{OUT} \times \left(V_{IN_MAX} - V_{OUT} \right)}{V_{IN_MAX} \times L \times f_{SW}}$$

  Vout / Vin_max can be simplified to the Duty Cycle, which at worst case at nominal inductance (highest voltage input of 15V) is equal to 0.33. Then, the numbers plug in to 0.33*(15-5)/(10 µH * 500000) = 0.67 A. As 0.67 A is about 33.5% of 2 A (the total output current), the worst case ripple current stays within the buck IC's recommended range of 20 to 60%.
- **Saturated Current:** The inductor's tolerance is listed as plus or minus 30% of 10 µH. So, at the worst inductance value (7 µH), the ΔI_L value works out to about 0.95 A, which means that the inductor needs to be able to handle a highest possible current of 2 A (maximum Iout per the onboarding guide) + 0.95 / 2 = 2.48 A. This inductor has a maximum saturated current of 4.4 A, so the inductor won't be affected even if it reaches extreme peak current.

- **Capacitors:** (The buck IC datasheet specifies what kind of capacitors we need and where, as well as roughly how many.)
  - The recommended capacitance value is already given by the buck IC datasheet, and it is about a value of Cout = 2x22 µF (2 capacitors in parallel, which can reduce output noise due to its placement), rated for 16 V for a 5 Vout value on the buck IC. However, that specific capacitor was out of a stock, and a similar one with the same capacitance is linked below that is rated for 10V. As this still exceeds the 50% margin for the capacitors as found in the Low Voltage guide, (5 V out * 1.5 = 7.5), this capacitor works as well. These two output capacitors are linked here: https://www.digikey.com/en/products/detail/samsung-electro-mechanics/CL31A226KPHNNNE/3886945, and the total price is $0.94.
  - Input capacitor linked here: https://www.digikey.com/en/products/detail/samsung-electro-mechanics/CL31B475KBHNNNE/3888447?s=N4IgTCBcDaIMIBkDMBGAQgFgOwFYDSaAEgHKkCiIAugL5A and costs $0.39 per unit. The buck IC data sheet requires greater than or equal to 4.7 µF of input capacitance, which is why we need this input capacitor.
  - Input high frequency capacitor and bootstrap capacitors (both served by the same component:

https://www.digikey.com/en/products/detail/kemet/C0603C104K5RACTU/1465594, and $0.28 total for both. Required by the buck IC datasheet and it specifies a 0.1 µF capacitor to filter out high frequency switching noise and the same magnitude capacitor to turn on the high-side switches.

- **Resistors:** 2 resistors are required per the buck IC datasheet, and they are at 100 kOhms (top) and 13.7 kOhms (bottom) respectively. According to the onboarding guide and the low voltage standards document, the resistors need to be sized at 0603. I picked the following resistors:
    - https://www.digikey.com/en/products/detail/yageo/AR0603FR-07100KL/5904594 for the 100kOhm resistor, at $0.52 per unit
    - https://www.digikey.com/en/products/detail/yageo/RC0603FR-0713K7L/726933?s=N4IgTCBcDaIEoGEAMA2JBmAYnAtEg7AIzoDS%2BAMiALoC%2BQA for the 13.7kOhm resistor, at $0.10 per unit.
- **The total price for all the components is roughly $4.59.**